

- 6** A sinusoidal voltage supply of peak voltage 8 V and period of 1.2 s is connected to a circuit as shown in Fig. 6.1. The circuit consists of four resistors P, Q, R and S, which has a resistance of $10\ \Omega$ each.

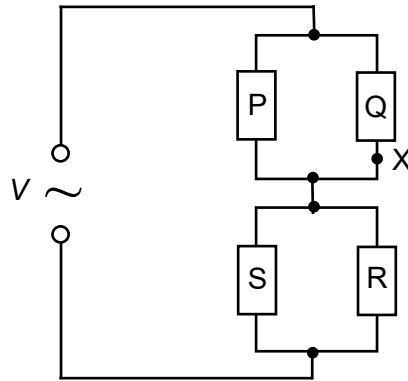


Fig. 6.1

- (a)** Calculate the maximum potential difference across resistor P.

potential difference = V [3]

Solution

$$\text{Total resistance} = \left(\frac{1}{10} + \frac{1}{10} \right)^{-1} + \left(\frac{1}{10} + \frac{1}{10} \right)^{-1} = 10\ \Omega$$

Using potential divider principle,

$$\begin{aligned} \text{Maximum potential difference across P} &= \frac{5}{10}(8) \\ &= 4\ \text{V} \end{aligned}$$

C1

M1

A1

- (b)** Determine the peak power dissipated across resistor P.

peak power = W [2]

Solution

$$\begin{aligned} \text{Peak power} &= \frac{V_o^2}{R_p} = \frac{4^2}{10} \text{ (ecf allowed)} \\ &= 1.6\ \text{W} \end{aligned}$$

M1

A1

(c)

An ideal diode is connected in series with resistor Q at point X.

On Fig. 6.2, sketch the variation with t of the p.d. across resistor Q for a time of 1.2 s. Add a scale to the y-axis.

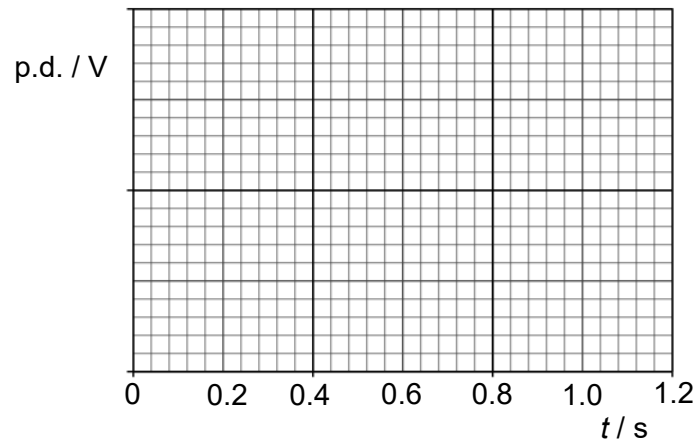
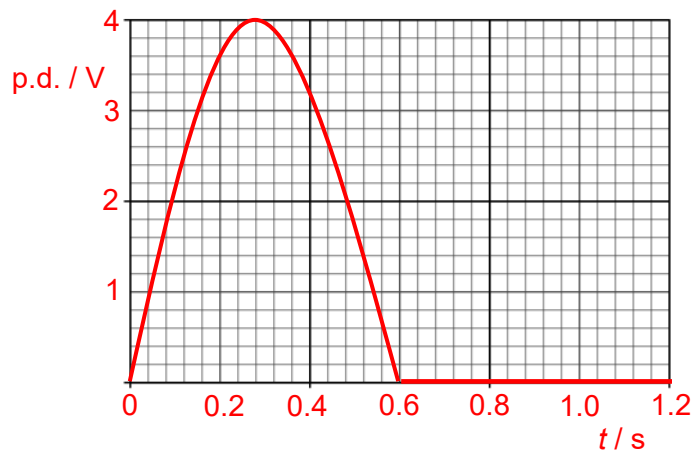


Fig. 6.2

[2]

Solution

Correct shape of graph. (accept cosine)
Correct labelling of graph.

B1
B1

[Total: 7]

